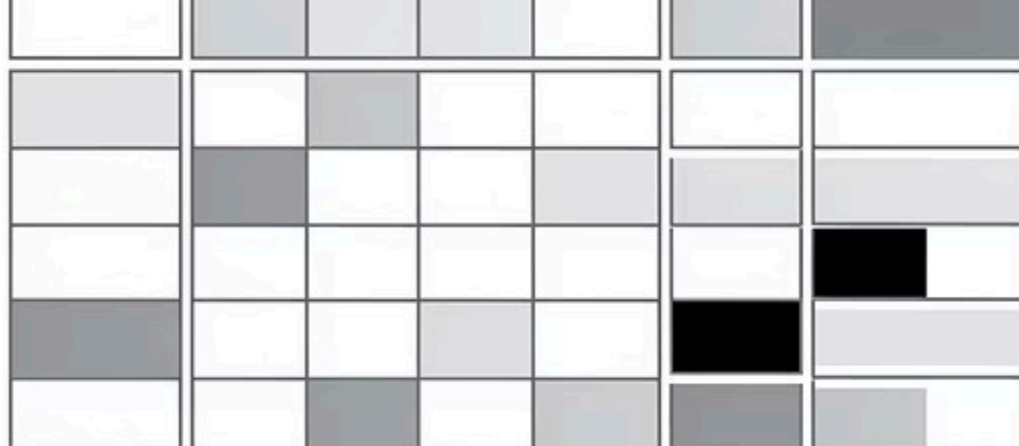




# Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • goal: uncover spending patterns, segments, product & subscription insights



|            |             |             |            |            |             |            |
|------------|-------------|-------------|------------|------------|-------------|------------|
| Light Gray | Medium Gray | Light Gray  | Light Gray | Light Gray | Medium Gray | Dark Gray  |
| Light Gray | White       | Medium Gray | White      | White      | White       | White      |
| White      | Dark Gray   | White       | White      | Light Gray | Light Gray  | Light Gray |
| White      | White       | White       | White      | White      | White       | Black      |
| Dark Gray  | White       | White       | Light Gray | White      | Black       | Light Gray |
| White      | White       | Medium Gray | White      | Light Gray | Medium Gray | Light Gray |
| White      | White       | Medium Gray | White      | Light Gray | Medium Gray | White      |

# Dataset at a Glance

Rows

3,900

Columns

18

Missing Data

37 values in Review Rating

# Key Features



## Demographics

Age, Gender, Location, Subscription



## Purchase Details

Item, Category, Amount, Season, Size, Color



## Behavior

Discounts, Promo codes, Frequency, Ratings, Shipping



# EDA — Python Workflow

## Load & Inspect

pandas: df.info() • .describe()

## Clean & Impute

Imputed review ratings by category median

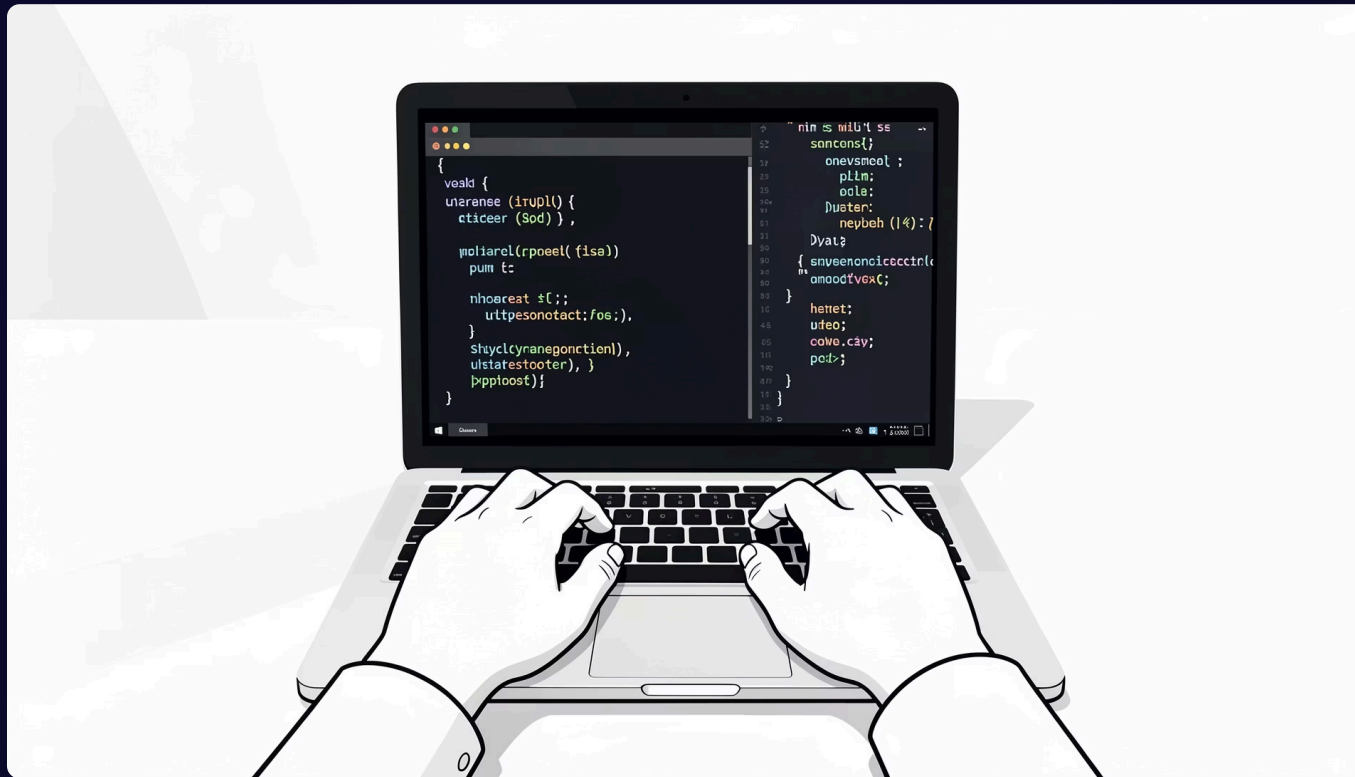
## Feature Eng.

age\_group, purchase\_frequency\_days

## DB Integration

Loaded cleaned dataframe into PostgreSQL

# Data Prep Details



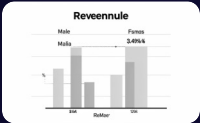
## Column Standardization

Renamed to snake\_case

## Redundancy Check

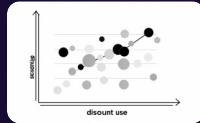
Dropped promo\_code\_used  
(redundant with discount\_applied)

# SQL Analysis — Selected Insights



## Revenue by Gender

Compare total revenue:  
male vs female



## High-Spending Discount Users

Users using discounts but  
spending above average

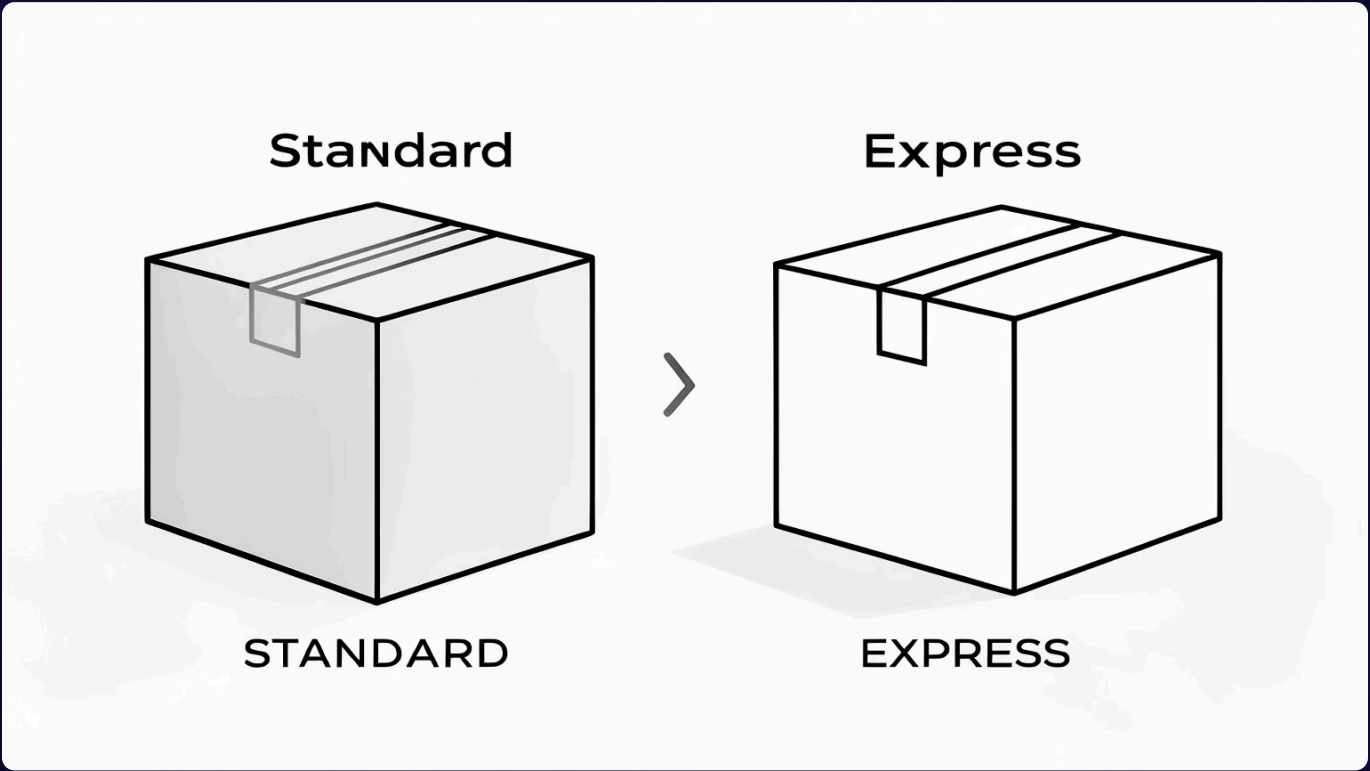
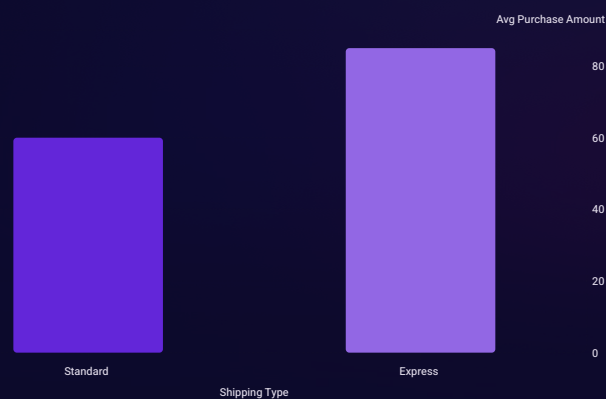


## Top 5 Products by Rating

Highest average review  
ratings

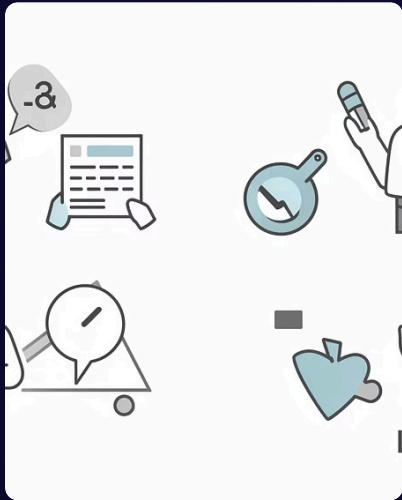
# More SQL Insights

Shipping Type Comparison: Express buyers spend more on average



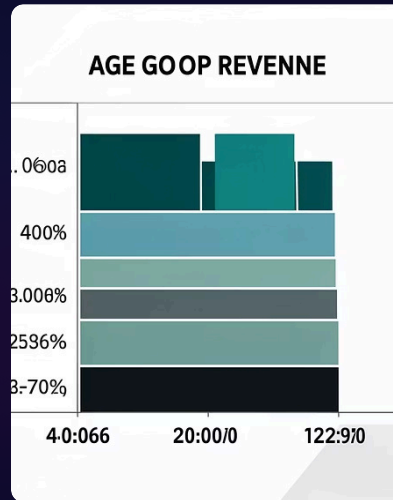


# Segmentation & Revenue



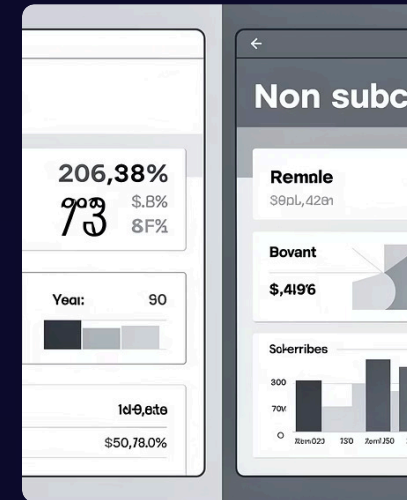
## Customer Segments

New • Returning • Loyal (based on purchase history)



## Revenue by Age Group

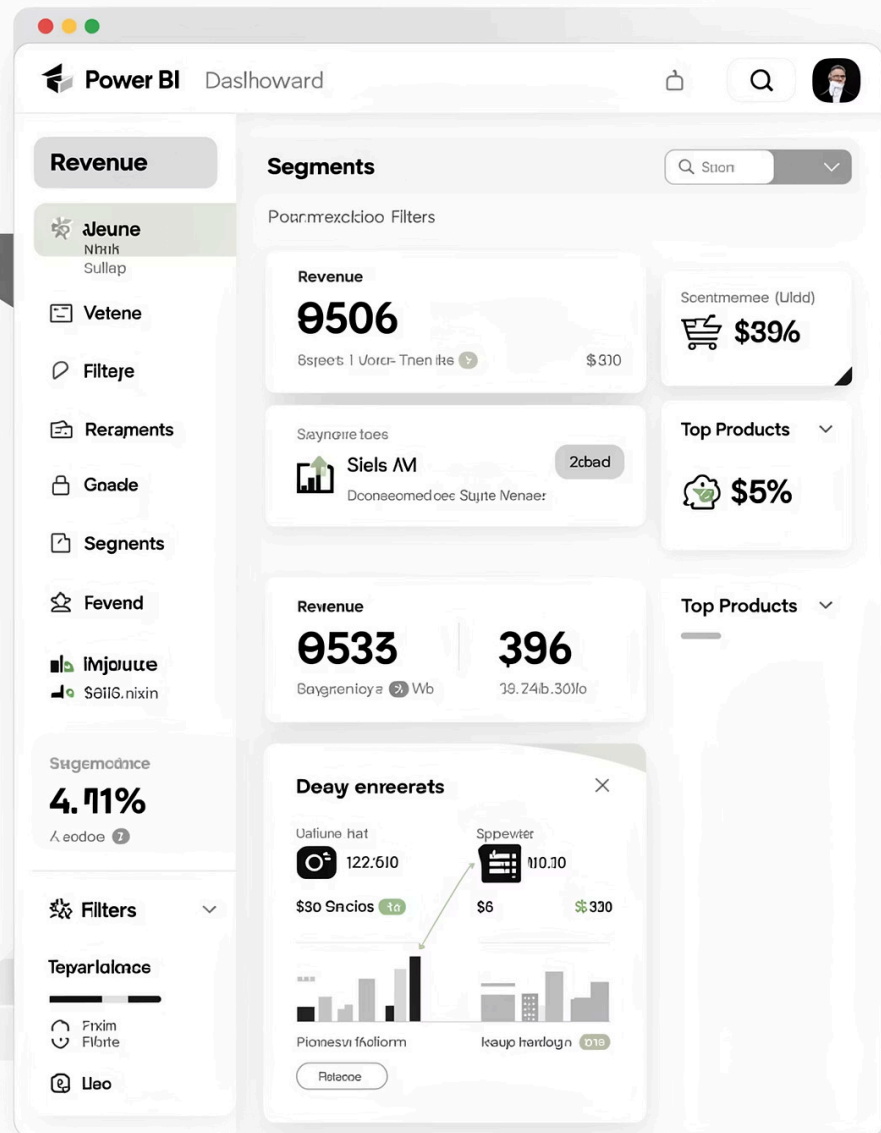
Age segments' total revenue contributions



## Subscribers vs Non-Subscribers

Compare avg spend and total revenue by subscription status





# Dashboard

Interactive Power BI dashboard built to present findings and explore segments

Supports drill-down by age, shipping, subscription, discounts

# Business Recommendations

## Increase Subscriptions

Promote premium value & exclusive benefits

## Strengthen Loyalty

Reward programs to convert repeat buyers into loyal customers

## Optimize Discounts

Reassess promotions to protect margins

## Refine Positioning

Feature top-rated, high-performing products in marketing

## Targeted Marketing

Focus on high-revenue age segments & express-shipping customers